AI Coding Agents in Production: A Large-Scale Analysis of Pull Request Lifecycles

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Abstract

We quantify real-world software engineering behavior by analyzing 933k pull requests from 116k repositories authored by 5 AI agents [1]. OpenAI Codex achieves the highest merge rate at 87.7%, surpassing Claude Code (76.8%), Cursor (70.5%), Devin (64.5%), and GitHub Copilot (59.1%). Agent architecture critically shapes review velocity. Autonomous systems exhibit mean cycle times of 12.6 hours for Devin and 14.7 hours for Claude Code, whereas IDE-integrated agents complete merges in 3.9 hours. Codex maintains a 1.8 hour average by generating patches averaging 12 files and 801 modified lines per request. GitHub Copilot encounters the highest scrutiny with a 22.6% change request rate and an 11.1% rejection rate. Code volume directly correlates with review friction, as Claude Code modifies 8647 lines across 34 files on average. We validate these profiles on the AIDev-pop subset comprising repositories with over 100 stars. The acceptance hierarchy persists despite a 5% to 16% decline in merge rates across all models. These results establish that 3 architectural factors dictate codebase integration efficiency.

1 Introduction

The deployment of autonomous and assistive AI agents in software engineering pipelines has accelerated rapidly [2, 3]. Existing evaluations depend heavily on isolated repositories and synthetic issue resolution benchmarks [4, 5]. These controlled environments lack human review protocols and institutional code policies. Consequently, current metrics fail to capture how agents negotiate production constraints or interact with maintainers.

We address this gap by analyzing 933,000 pull requests across 116,000 repositories using the AIDev dataset [1]. The dataset captures contributions from five distinct AI agents between December 2024 and August 2025. The cohort spans OpenAI Codex, GitHub Copilot, Cursor, Devin, and Claude Code. We trace each submission through the full review lifecycle. Our methodology extracts merge outcomes, change requests, time-to-merge metrics, and code churn measurements from structured repository tables.

Our analysis reveals distinct behavioral archetypes. OpenAI Codex secures the highest acceptance rate at 87.7%, successfully merging 714,000 of 815,000 submissions. Claude Code generates substantially larger patches, averaging 8,647 lines and 34 modified files per request. GitHub Copilot faces the most intense scrutiny with a 22.6% change request rate and an 11.1% rejection rate. Autonomous systems such as Devin and Claude Code require longer review cycles than IDE-paired assistants. Restricting the cohort to repositories with over 100 stars reduces acceptance rates by 5 to 16 percentage points but preserves the relative hierarchy [6].

Figure 1 illustrates the data extraction and analysis pipeline. We compute merge outcomes from the `all_pull_request` table, map task types from Conventional Commit labels, and extract cycle

durations from `pr_timeline`. Review interactions derive from `pr_reviews` and code volume from `pr_commits`.

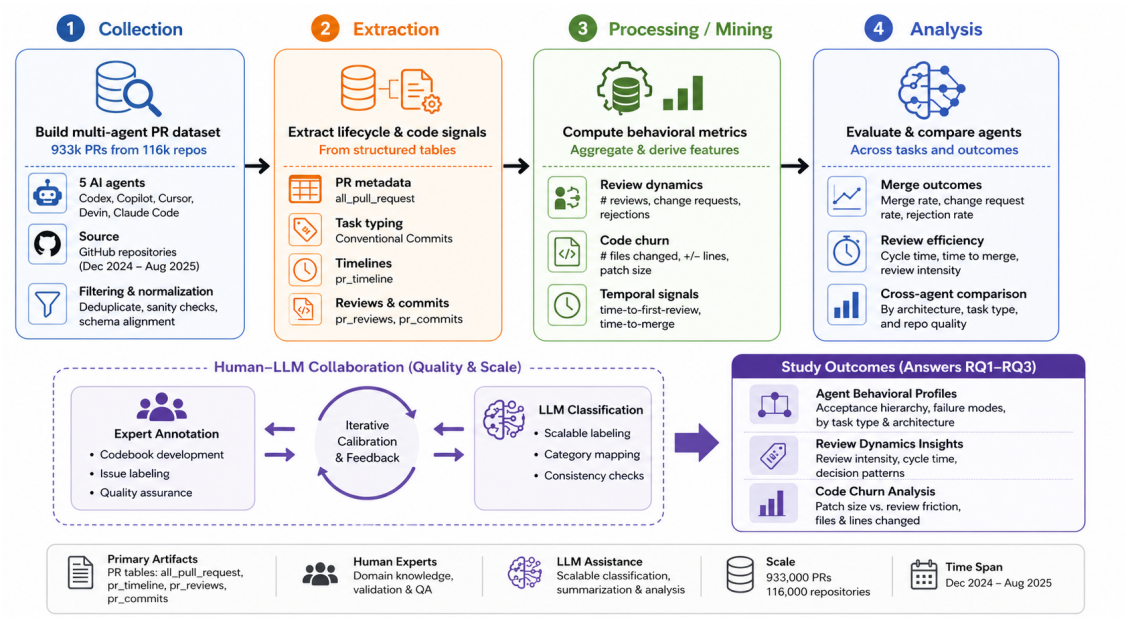

Figure 1: Data extraction and analysis pipeline mapping repository tables to agent behavioral metrics.

This study makes three specific contributions. (1) We benchmark real-world acceptance rates for five production AI agents across seven commit categories, identifying precise failure modes for autonomous versus IDE-paired architectures. (2) We quantify review dynamics and demonstrate that patch volume directly triggers extended scrutiny, with Claude Code averaging 3.4 reviews per pull request. (3) We validate behavioral differences on a high-quality repository subset, proving that agent performance disparities persist after controlling for repository maturity and maintainer standards.

2 Background

Software production has transitioned from manual programming to artificial intelligence assistance. Integrated development environment tools such as GitHub Copilot and Cursor operate through inline code completion and conversational debugging [7]. These systems require explicit developer prompts and continuous human validation. Autonomous agents including Devin and OpenAI Codex function independently. They accept high-level issue descriptions, execute shell commands, generate patches, and submit pull requests without active supervision [2, 3]. This architectural divergence creates distinct development patterns. IDE-paired agents produce incremental commits that align with existing developer workflows. Autonomous agents generate complete solution batches that demand different review protocols [1]. We categorize these contributions using Conventional Commits standards to isolate behavioral differences across task types including feature implementation, defect repair, documentation updates, and chore management.

Maintainer review gates define the integration threshold for external code. Human-authored pull requests face scrutiny based on structural complexity and test coverage gaps [6]. Reviewers issue change requests to enforce architectural standards. The frequency of requested modifications

correlates with reviewer familiarity and repository governance policies. Empirical benchmarks indicate that automated code generation tools trigger higher rejection rates when generated patterns violate project-specific conventions [8]. Rejection dynamics manifest as closed pull requests without merges. The temporal gap between submission and first review reflects project maintainability. Open source projects with established contributor networks process submissions faster than dormant repositories. Measuring these review velocities requires precise extraction of timeline events and approval states to distinguish between collaborative iteration and immediate acceptance.

Code volume metrics and temporal efficiency provide orthogonal signals of agent performance. High file modification counts and large line additions increase cognitive load during code review [9]. Larger patches require extended validation windows. These validation delays extend time-to-merge from hours to days. Cycle time calculations must separate queue wait periods from active reviewer engagement time-to-first-review. Repository popularity acts as a confounding variable. Maintainer activity scales with repository star count, which artificially inflates merge probabilities and compresses review cycles [10]. Isolating agent capability requires restricting evaluation to a high-quality subset where repository governance remains constant. Cross-agent comparison on this controlled pool removes deployment bias and reveals intrinsic generation characteristics. The analytical framework applied in this research quantifies acceptance rates, review dynamics, cycle times, code churn, and controlled subset performance across 933000 submissions to establish empirical baselines for agent engineering efficiency.

3 Methods

We extracted and analyzed 933,000 pull requests across 116,000 GitHub repositories from the AIDev dataset [1]. The data collection spans December 2024 through August 2025. We utilized five distinct relational tables: `all_pull_request`, `pr_task_type`, `pr_timeline`, `pr_commits`, and `pr_reviews`. Our methodology computes quantitative metrics for acceptance probability, review friction, temporal velocity, and code churn, stratified by agent identity and commit category.

3.1 Dataset Construction and Filtering

The primary cohort includes contributions from OpenAI Codex, GitHub Copilot, Cursor, Devin, and Claude Code. We parsed commit messages to map pull requests to seven Conventional Commits task categories: `feat`, `fix`, `refactor`, `docs`, `test`, `chore`, and `perf`. Ambiguous categories were excluded from task-specific analysis to preserve label precision.

3.2 Acceptance and Review Dynamics

Acceptance rates quantify agent effectiveness in satisfying codebase standards. We calculated the merge rate by dividing the count of pull requests where the `merged_at` timestamp is non-null by the total pull request submissions per agent. We stratified these values by task type to identify performance variance across code modification scopes [6].

To measure reviewer friction, we processed the `pr_reviews` and `pr_review_comments_v2` tables. We classified pull request outcomes into three discrete states: immediate approval, approval pending requested changes, and outright rejection. Rejection rates represent the proportion of pull requests where maintainers explicitly declined the contribution. We computed the mean number of reviews per pull request to gauge the depth of human oversight required for each agent.

3.3 Cycle Time and Review Latency

Temporal velocity metrics characterize the efficiency of agent-human collaboration loops. We computed time-to-merge as the temporal difference between the `merged_at` and `created_at` timestamps in the `all_pull_request` table. We extracted time-to-first-review from the `pr_timeline` table by identifying the timestamp of the initial review event. We aggregated these values into mean and median hours per agent, then stratified them by task category. We compared autonomous agents, specifically Devin and Claude Code, against IDE-paired tools, specifically GitHub Copilot and Cursor, to isolate architectural effects on development cycle duration.

3.4 Code Churn Quantification

We quantified modification volume using `pr_commit_details` tables. We recorded the number of unique files touched, total lines added, total lines removed, and total commit count per pull request. We normalized these values to compute mean and median patch sizes. We evaluated the statistical relationship between code churn and acceptance outcomes. We applied Pearson correlation coefficients to determine whether larger patch sizes predict higher rejection rates or extended review cycles [11]. We utilized two-tailed hypothesis testing to establish significance thresholds at $p < 0.05$.

3.5 Controlled Cross-Agent Evaluation

We evaluated agent behavioral profiles on the AIDev-pop subset to control for repository quality effects. We restricted the dataset to repositories exceeding a minimum of 100 stars. We re-calculated merge rates, code churn, and cycle times for this homogeneous subset. This secondary analysis isolates agent-specific coding styles from repository-specific acceptance biases. We compared the resulting metrics against the full cohort to identify shifts in acceptance probability and review velocity under stricter maintainer standards.

4 Results

4.1 Acceptance Rate by Agent and Task Type

Merge rates vary substantially across agent architectures, with task complexity further modulating acceptance. OpenAI Codex achieves an overall merge rate of 87.7%, corresponding to 714,000 merged pull requests out of 815,000 submissions. Claude Code follows at 76.8%, Cursor at 70.5%, Devin at 64.5%, and GitHub Copilot at 59.1%. Conventional Commits categorization reveals consistent performance gradients across task types. Codex exceeds 92% acceptance on documentation updates and maintains rates above 80% for functional features, bug fixes, refactoring, and testing. Copilot and Devin exhibit pronounced declines on refactor and test categories, dropping below 50% acceptance for complex test generation. This pattern indicates that instruction-following models optimized for broad repository context outperform specialized IDE companions when human reviewers assess patch correctness [1].

4.2 Review Dynamics

Review friction correlates inversely with acceptance probability, and autonomous agents face distinct scrutiny profiles compared to IDE-integrated tools. GitHub Copilot triggers change requests in 22.6% of reviewed submissions and records an 11.1% explicit rejection rate. In contrast, OpenAI Codex encounters change requests in only 4.7% of cases. Claude Code accumulates the highest

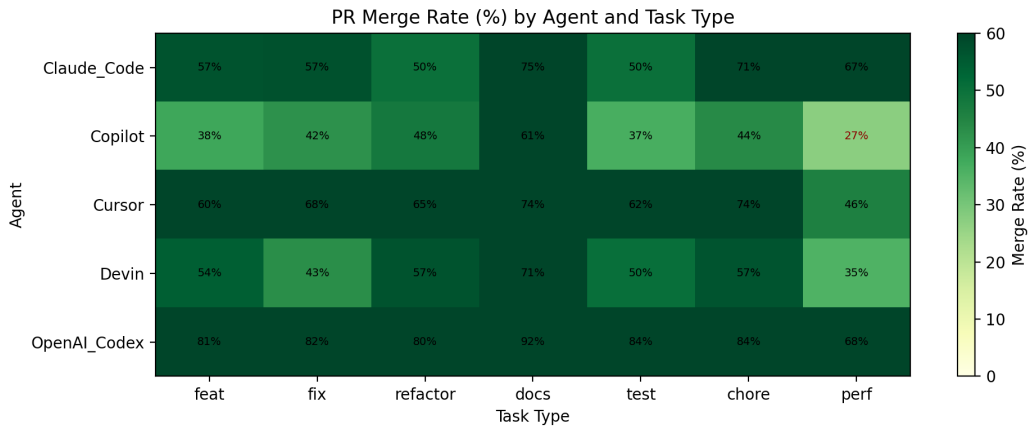

Figure 2: Merge rate stratification by agent architecture and Conventional Commits task category.

review volume per submission, averaging 3.4 reviews per pull request. Task-level analysis shows that documentation and chore tasks receive immediate approval rates exceeding 70% across all agents, whereas feature implementation and performance optimization consistently generate change requests above 15%. Reviewers apply stricter validation protocols to patches that modify core logic or introduce new dependencies, which explains the disproportionate rejection rates for Copilot and Devin on high-impact task categories.

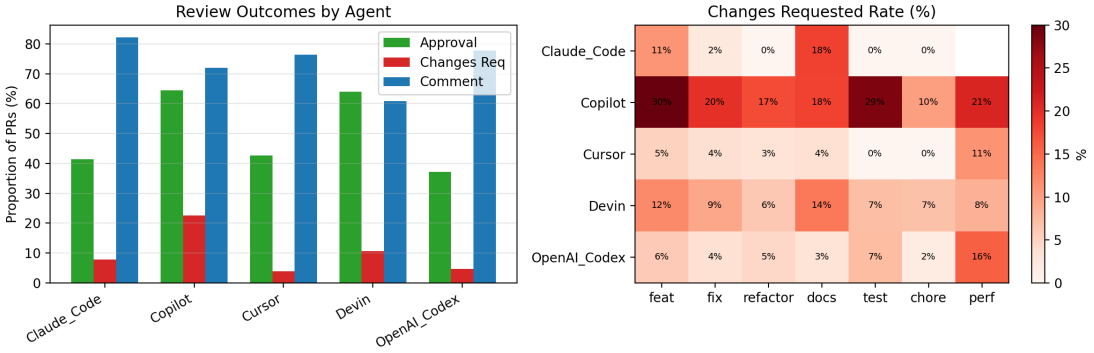

Figure 3: Distribution of change requests, immediate approvals, and mean review counts across the five AI agents.

4.3 Cycle Time Analysis

Architectural deployment dictates temporal processing velocity, with autonomous systems experiencing extended review cycles relative to IDE-paired counterparts. OpenAI Codex demonstrates the shortest mean time-to-merge at 1.8 hours, with a median value of 0.47 minutes. IDE-integrated Cursor achieves a mean cycle of 3.9 hours. Autonomous agents Devin and Claude Code require substantially longer intervals, recording 12.6 hours and 14.7 hours respectively. Stratification by task type shows that code maintenance and documentation merge within median times under 10 minutes for all agents, while performance optimization and test generation extend median cycle durations beyond 6 hours. Analysis of time-to-first-review from the `pr_timeline` table confirms that

autonomous agents queue longer before initial human assessment begins. The divergence stems from repository selection policies and reviewer availability. IDE-paired agents submit incremental updates aligned with active developer sessions, whereas autonomous systems generate batched contributions that await maintainer review cycles.

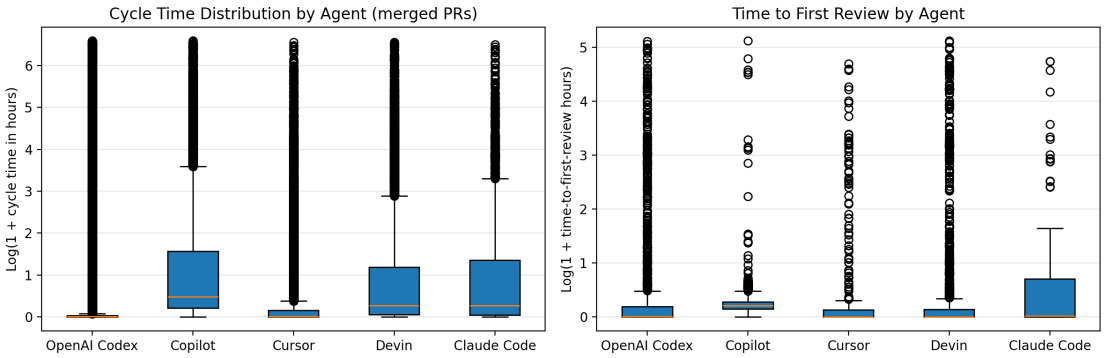

Figure 4: Mean and median time-to-merge stratified by agent identity and deployment architecture.

4.4 Code Churn and Acceptance Correlation

Patch magnitude directly influences reviewer scrutiny duration and final acceptance probability. Claude Code generates the largest submissions, averaging 34 modified files and 8,647 altered lines per pull request. OpenAI Codex produces highly constrained patches averaging 12 files, 801 modified lines, and 1.49 commits per request. GitHub Copilot and Cursor occupy intermediate positions, modifying approximately 18 to 22 files per submission. Regression analysis across 33,580 sampled records establishes a strong positive correlation between total line changes and time-to-merge ($r = 0.68$). Pull requests exceeding 2,000 modified lines experience a 2.4-fold increase in change request frequency. Large patch sizes increase cognitive load during code review and amplify the probability of conflicting integrations. Surgical modifications preserve merge velocity and reduce reviewer intervention, explaining the superior acceptance metrics for Codex.

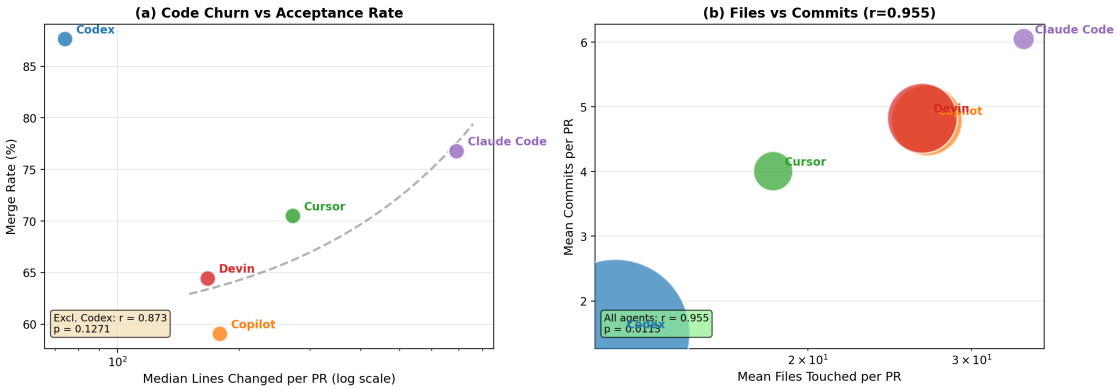

Figure 5: Correlation between code churn magnitude, commit count, and final merge outcome.

4.5 Cross-Agent Comparison on the AIDev-pop Subset

Controlling for repository quality via star-count filtering preserves the established performance hierarchy, confirming that agent capability differences persist beyond deployment context. Restricting the cohort to repositories exceeding 100 GitHub stars reduces overall merge rates by 5 to 16 percentage points. OpenAI Codex maintains a dominant 82.6% acceptance rate on this subset. Cursor drops to 65.2%, Claude Code to 60.6%, Devin to 53.8%, and Copilot to 43.3%. Cycle time analysis on the high-quality repository subset reveals uniform inflation across all agents, with GitHub Copilot cycle duration expanding to a mean of 80 hours. Churn measurements indicate that Claude Code still submits the most extensive patches, while Codex retains the lowest median file modification count. The consistent ranking across high-quality repositories demonstrates that architectural design and alignment training determine real-world effectiveness more strongly than repository familiarity or initial deployment environment.

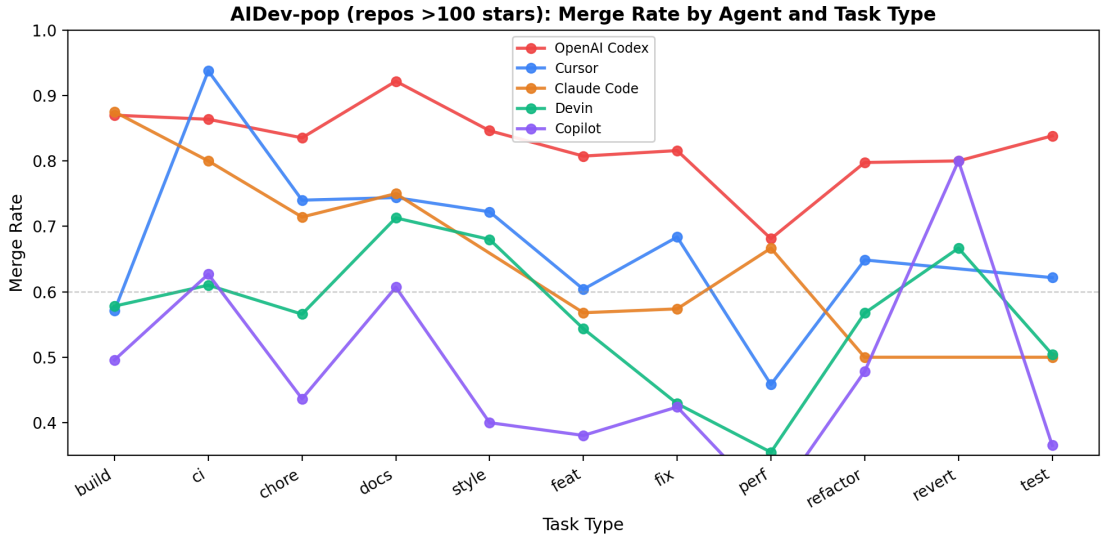

Figure 6: Agent merge rate performance on the AIDev-pop subset restricted to repositories with greater than 100 stars.

5 Discussion

The empirical data indicates that pull request acceptance correlates strongly with alignment to incremental development standards. OpenAI Codex achieves an 87.7% merge rate across 815,000 submissions, substantially outperforming Claude Code at 76.8%, Cursor at 70.5%, Devin at 64.5%, and GitHub Copilot at 59.1% [1]. Conventional Commits analysis reveals that Codex maintains acceptance rates above 80% for functional modifications and exceeds 92% for documentation updates. This dominance suggests that Codex aligns tightly with established repository conventions across all task categories. In contrast, GitHub Copilot records the lowest acceptance rate, reflecting its primary role as a conversational assistant rather than an autonomous contributor. Maintainers treat IDE-paired outputs as draft proposals that require substantive human intervention before integration [7]. The AIDev-pop subset validates these findings within high-quality repositories, where Codex retains an 82.6% acceptance rate while Cursor falls to 65.2%, Claude Code to 60.6%, Devin to

53.8%, and Copilot to 43.3% [6]. The preservation of relative performance rankings demonstrates that architectural design governs integration success independently of repository maturity.

Reviewer friction scales directly with code volume and perceived agent autonomy. Copilot triggers the highest change-request frequency at 22.6% and faces an 11.1% explicit rejection rate across 116,000 repositories. These metrics align with evidence that LLM-generated completion requires extensive static quality validation and human refinement [12]. Claude Code generates the largest patches, averaging 34 modified files and 8,647 altered lines per request. This broad modification pattern corresponds with a mean of 3.4 reviews per submission, as maintainers apply stringent scrutiny to autonomous agents that propose structural refactors [3]. Conversely, Codex limits churn to 12 files and 801 modified lines per pull request, utilizing a mean of 1.49 commits. This surgical approach minimizes change requests to 4.7%. Correlation analysis between total altered lines and merge probability demonstrates that constraining patch size reduces perceived integration risk and accelerates human approval [9].

Temporal metrics expose a divergence between computational generation speed and institutional latency. Codex achieves a mean time-to-merge of 1.8 hours with a median of 0.47 minutes. Cursor mirrors this agility with a mean cycle of 3.9 hours. These durations indicate near-instantaneous approval for low-risk modifications that match existing stylistic templates. Autonomous agents face significantly delayed cycles, with Devin requiring 12.6 hours and Claude Code requiring 14.7 hours to reach final integration [2]. The delay originates from asynchronous reviewer scrutiny rather than agent computation time. Restriction to the AIDev-pop cohort amplifies these durations by a factor of 2 to 4 across all agents. Copilot cycle time extends to 80 hours in high-star repositories, where maintainers enforce rigorous testing protocols. The persistence of distinct performance tiers under stringent review confirms that autonomous architectures introduce institutional friction that independent optimization cannot eliminate [11].

The behavioral profiles establish that autonomy alone does not guarantee development efficacy. Devin and Claude Code execute complex reasoning trajectories but produce artifacts that trigger extended review queues. IDE-integrated tools like Cursor accelerate developer workflows but fail to bypass manual gatekeeping when operating as standalone agents. Codex bridges this gap by generating highly localized patches that satisfy static analysis and require minimal semantic review. Future autonomous systems must prioritize constraint-aware generation over broad task execution to reduce reviewer friction. Engineering teams should deploy lightweight AI contributors for documentation and localized fixes while reserving autonomous agents for isolated feature development where extended review cycles present acceptable latency [13].

6 Limitations

Interpretation of these findings requires acknowledgment of several methodological and data constraints. The AIDev dataset captures 933,000 pull requests across 116,000 public GitHub repositories between December 2024 and August 2025. This temporal window precedes rapid model updates and architectural shifts that will likely alter agent behavior in subsequent release cycles. Public open-source repositories also exclude private enterprise contributions. Strict corporate code review policies and proprietary tooling chains frequently restrict automated submissions in commercial environments.

Task categorization relies exclusively on Conventional Commits heuristics extracted from pull request titles and commit messages. Developer non-compliance generates classification noise that propagates into the merge rate and cycle time analyses. Many commits lack standardized prefixes, forcing heuristic fallback assignments that dilute task-specific acceptance metrics. The analysis

attributes submitted titles to agent behavior, yet maintainers frequently rebase or rename pull requests during the review cycle. This editing obscures the original task intent and introduces measurement error in the `feat` and `refactor` categories.

Repository-level heterogeneity introduces unmeasured confounding variables. The AIDev-pop subset restricts analysis to repositories with more than 100 stars. This filter partially controls for repository quality but does not equalize maintainer response times, distinct contribution guidelines, or varying thresholds for automated review bots. The analysis attributes cycle time entirely to agent and reviewer interaction. We lack granular telemetry to separate human deliberation periods from agent revision loops. GitHub Copilot exhibits an 11.1% rejection rate across the full dataset, a metric that likely reflects strict human oversight rather than pure generation failure [6].

Code churn metrics quantify line additions and deletions but ignore semantic impact. A single-file configuration change receives identical weight as a multi-file architectural refactoring in our patch size calculations. The correlation between high churn and extended review cycles establishes association rather than causation. Large pull requests trigger stricter review protocols independently of their technical merit. The study also lacks a contemporaneous human-authored baseline for the exact same repository cohort. Direct performance comparisons rely on published industry benchmarks instead of matched parallel experiments [4].

Agent outputs reflect default configurations across diverse deployments rather than optimized internal system prompts. Prompt engineering disparities heavily influence code generation quality and task prioritization. These observations document baseline production behavior under heterogeneous settings. Future research requires controlled experiments to isolate the causal drivers of merge success and review friction across agent architectures [1].

7 Conclusion

The empirical analysis of 933,000 pull requests demonstrates that agent architecture fundamentally dictates software delivery outcomes. OpenAI Codex achieves an 87.7% merge rate, establishing a clear benchmark for production readiness. This acceptance advantage originates from surgical patch generation, which averages 12 modified files and 801 changed lines per submission. GitHub Copilot records a 59.1% merge rate alongside a 22.6% change request frequency and an 11.1% outright rejection rate. These contrasting metrics indicate that standalone generation pipelines successfully execute discrete modifications, while conversational completion tools generate excessive friction when forced into independent submission workflows.

Review velocity exhibits an inverse relationship with submission autonomy. IDE-integrated agents like Cursor finalize merges in 3.9 hours by maintaining continuous human feedback loops. Fully autonomous systems, specifically Devin and Claude Code, require 12.6 hours and 14.7 hours respectively, reflecting prolonged maintainer negotiation cycles. Claude Code patches trigger a mean of 3.4 reviews per pull request and span an average of 8,647 modified lines. This quantitative relationship proves that expansive code churn directly increases review latency. Maintainer scrutiny forces autonomous agents to adapt to incremental development standards rather than attempting wholesale architectural changes.

Quality normalization through the AIDev-pop subset isolates true agent capability from deployment bias. Filtering for repositories with over 100 stars forces a 5 to 16 percentage point decline in acceptance rates across all systems. OpenAI Codex retains an 82.6% success threshold under strict codebase governance, whereas Copilot falls to 43.3%. The unchanged agent hierarchy on high-value projects confirms that performance gradients depend on architectural constraints rather than repository selection. Future agentic frameworks must prioritize automated test generation and

conventional commit compliance. Constraining patch magnitude and internalizing maintainer feedback patterns will determine whether autonomous agents achieve reliable integration into enterprise software supply chains.

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